97 b) for a fluence of 10^{25} n m^{-2} (a) assuming no consumption of ^{235}U formed, (b) taking ^{235}U fission and capture into account.

21.5. Explain why ^{228}Th and ^{232}U is a nuisance in the thorium-uranium fuel cycle.

21.6. A reactor starting with 3% ^{235}U produces 6000 MWd energy/t U fuel each year. Neglecting fission in ^{238}U, (a) how much fission products have been produced after 5 years? (b) What is the ^{235}U concentration if plutonium fission also is taken into account?

21.7. In the example above, 1 t U as fuel elements is removed from the reactor after 2 years. Using Fig. 21.7, (a) what is the total radioactivity from the fission products after 1 y cooling time? (b) Which FP elements are the most radioactive ones at this time?

21.8. Calculate the decontamination factor required for (a) fission product activity, and (b) for gadolinium in commercial plutonium nitrate produced from PWR fuel (Tables 21.2 and 21.7) at t_{cool} 10 y.

21.9. 0.0015 Ci ^{239}Pu is released annually from a reprocessing plant. What will be the corresponding release of ^{238}Pu and ^{240}Pu for typical isotopic plutonium composition of LWR fuel?

21.10. Calculate the natural radiotoxicity value In_w of 1 km^3 of land (density 2 600 kg m^{-3}) containing 3 weight ppm ^{238}U with daughter products. Only ^{226}Ra has to be considered.

21.11. A tank contains 100 m^3 5 y old HLLW. Analyses show that a 1 ml sample contains 1.09 GBq of ^{90}Sr, which is the only Sr activity. (a) Calculate the heat production for a waste of composition in Table 21.9 left column. (b) How many 500 kg glass cylinders would be needed (assume the glass contains 10% FPs) to contain all the solidified waste? (c) How many 1000 MW_e PWR reactor years does this waste correspond to?

21.12. Assume that Ru, Rh, and Pd are recovered from the waste from a 10 GW_e program. What will the annual amounts and specific radioactivities be at $t_{cool} =$ 10 y for each of them?

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CHAPTER 22

Behavior of Radionuclides in the Environment

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The main objection against nuclear power is the risk of spread of "radioactivity" (radioactive elements) to the environment where it may cause health effects in humans. Such effects have already been discussed in chapter 15. As mentioned previously there are several suggested methods for the treatment of used nuclear fuel. However, all of them will require a final repository for the waste. The main difference is the total storage time, the different barriers, and the radiotoxicity of the final waste. In any case, if the repository is breached and ground water comes in contact with the fuel the readionuclides will be dissolved to a smaller or greater extent, depending on element and form. In this chapter, the chemical aspects of the sources of releases, and of the migration of the radionuclides in the environment, will be discussed. Their chemical properties, together with hydrology, determine how fast they will move from their point of entry into the

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